Page 1: Operating System Fundamentals

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An operating system manages hardware and provides services to applications. It controls the CPU, memory, storage, devices, files, security, and user interaction.

Key points

• Understand the definition and purpose of the concept.

• Identify the major components, algorithms, or conditions involved.

• Compare related approaches and understand their practical trade-offs.

• Use this section to test document retrieval, chunking, embeddings, and question answering in the IntelliLearn RAG pipeline.

Page 2: Processes and Threads

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A process is a program in execution. Process state, PCB, context switching, process creation, and termination are fundamental OS concepts. Threads provide smaller units of execution within a process and can share process resources.

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Page 3: CPU Scheduling

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CPU scheduling selects a ready process for execution. FCFS, SJF, SRTF, Priority Scheduling, and Round Robin have different trade-offs involving waiting time, turnaround time, response time, throughput, and fairness.

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Page 4: Process Synchronization

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Concurrent processes can create race conditions when they access shared data. Critical sections, mutexes, semaphores, monitors, and synchronization protocols help ensure correct execution.

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Page 5: Deadlocks

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A deadlock occurs when processes wait indefinitely for resources held by other processes. Mutual exclusion, hold and wait, no preemption, and circular wait are the four necessary conditions.

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• Identify the major components, algorithms, or conditions involved.

• Compare related approaches and understand their practical trade-offs.

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